

Taxi Booking System:

```
import java.util.*;
```

```
class Taxi {
    private int taxild;
    private String driverName;
    private String currentLocation;
    private double totalEarnings;
    private ArrayList<String> bookingHistory = new ArrayList<>();

    public Taxi(int taxild, String driverName, String currentLocation){
        this.taxild=taxild;
        this.driverName=driverName;
        this.currentLocation=currentLocation.toUpperCase();
    }
    public int getTaxild(){ return taxild; }
    public String getDriverName(){ return driverName; }
    public String getCurrentLocation(){ return currentLocation; }
    public void setCurrentLocation(String l){ currentLocation=l.toUpperCase(); }
    public double getTotalEarnings(){ return totalEarnings; }
    public void setTotalEarnings(double e){ totalEarnings=e; }
    public ArrayList<String> getBookingHistory(){ return bookingHistory; }
    public void addBooking(String b){ bookingHistory.add(b); }
}
```

```
class TaxiBookingSystem{
    private ArrayList<Taxi> taxis=new ArrayList<>();

    public void registerTaxi(Scanner sc){
        System.out.print("Enter Taxi ID : ");
        int id=sc.nextInt(); sc.nextLine();
        System.out.print("Enter Driver Name : ");
        String name=sc.nextLine();
        System.out.print("Enter Current Location (A/B/C/D/E): ");
        String loc=sc.nextLine();
        taxis.add(new Taxi(id,name,loc));
        System.out.println("Taxi Registered Successfully!");
    }
}
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public void viewTaxis(){
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if(taxis.isEmpty()){ System.out.println("No taxis registered."); return; }
for(Taxi t:taxis){
    System.out.println("-----");
    System.out.println("Taxi ID : "+t.getTaxiId());
    System.out.println("Driver : "+t.getDriverName());
    System.out.println("Location : "+t.getCurrentLocation());
    System.out.println("Earnings : ₹"+t.getTotalEarnings());
}
}

private int pos(String l){
    switch(l.toUpperCase()){
        case "A": return 0;
        case "B": return 15;
        case "C": return 30;
        case "D": return 45;
        case "E": return 60;
    }
    return -1;
}

public void bookTaxi(Scanner sc){
    if(taxis.isEmpty()){ System.out.println("No taxis registered."); return; }

    System.out.print("Pickup Location (A/B/C/D/E): ");
    String pickup=sc.nextLine().toUpperCase();

    ArrayList<Taxi> available=new ArrayList<>();
    System.out.println("Available Taxis:");
    for(Taxi t:taxis){
        if(t.getCurrentLocation().equalsIgnoreCase(pickup)){
            available.add(t);
            System.out.println("Taxi ID: "+t.getTaxiId()+" Driver: "+t.getDriverName());
        }
    }
    if(available.isEmpty()){
        System.out.println("No Taxi Available.");
        return;
    }

    System.out.print("Choose Taxi ID: ");

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int id=sc.nextInt(); sc.nextLine();

Taxi taxi=null;
for(Taxi t:available){
    if(t.getTaxiId()==id){ taxi=t; break; }
}
if(taxi==null){
    System.out.println("Invalid Taxi ID");
    return;
}

System.out.print("Customer Name: ");
String customer=sc.nextLine();
System.out.print("Drop Location (A/B/C/D/E): ");
String drop=sc.nextLine().toUpperCase();

int distance=Math.abs(pos(drop)-pos(pickup));
if(distance<0){ System.out.println("Invalid Location"); return; }

double fare;
if(distance==0) fare=100;
else if(distance==15) fare=200;
else if(distance==30) fare=350;
else if(distance==45) fare=500;
else fare=650;

System.out.println("Distance : "+distance+" KM");
System.out.println("Fare : ₹"+fare);

taxi.setCurrentLocation(drop);
taxi.setTotalEarnings(taxi.getTotalEarnings()+fare);
taxi.addBooking("Customer:"+customer+" Pickup:"+pickup+" Drop:"+drop+"
Distance:"+distance+"KM Fare:₹"+fare);
    System.out.println("Booking Successful!");
}

public void displayBookingHistory(){
    for(Taxi t:taxis){
        System.out.println("Taxi "+t.getTaxiId());
        if(t.getBookingHistory().isEmpty()) System.out.println("No Bookings");
        else for(String s:t.getBookingHistory()) System.out.println(s);
    }
}

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    }
}

public void displayEarnings(Scanner sc){
    System.out.print("Enter Taxi ID: ");
    int id=sc.nextInt();
    for(Taxi t:taxis){
        if(t.getTaxiId()==id){
            System.out.println("Driver: "+t.getDriverName());
            System.out.println("Earnings: ₹"+t.getTotalEarnings());
            return;
        }
    }
    System.out.println("Taxi Not Found");
}

public void totalTaxis(){
    System.out.println("Total Registered Taxis : "+taxis.size());
}
}

public class Mainn{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        TaxiBookingSystem system=new TaxiBookingSystem();

        while(true){
            System.out.println("\n===== Taxi Booking System =====");
            System.out.println("1.Register Taxi");
            System.out.println("2.View Taxis");
            System.out.println("3.Book Taxi");
            System.out.println("4.Booking History");
            System.out.println("5.Display Earnings");
            System.out.println("6.Total Taxis");
            System.out.println("7.Exit");
            System.out.print("Enter Choice: ");
            int ch=sc.nextInt();
            sc.nextLine();

            switch(ch){
                case 1: system.registerTaxi(sc); break;

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case 2: system.viewTaxis(); break;
case 3: system.bookTaxi(sc); break;
case 4: system.displayBookingHistory(); break;
case 5: system.displayEarnings(sc); break;
case 6: system.totalTaxis(); break;
case 7: System.out.println("Thank You"); sc.close(); return;
default: System.out.println("Invalid Choice");
    }
    }
    }
}
```